BORONG XU

Research Assistant at SLAI

+852 8491 8005

borongxu@outlook.com

boring180-github-io.vercel.app/

borong-xu-52829a293

boring180

ABOUT ME

Research Assistant at the Shenzhen Loop Area Institute (SLAI) and a top 3% graduate of HKUST's Computer Engineering (CPEG) program. Passionate about robotics and embodied intelligence, with research experience spanning visual-inertial odometry, SLAM, multi-camera localization, and quadruped control, as well as deep learning, vision transformers, and LLM-based agent systems. Eager to push the boundaries of autonomous systems in real-world deployments.

EDUCATION

The Hong Kong University of Science and Technology (HKUST) — *Hong Kong SAR*

Bachelor of Engineering in Computer Engineering, First Class Honors — *Graduated Jun 2026*

- GPA 3.84/4.3 (Top 3% among 157 students)
- Major GPA 4.03/4.3

High School Affiliated to Renmin University of China (RDFZ) — *Beijing, China*

High School Diploma(GaoKao) — *Graduated Jun 2022*

- Activities:
  - FIRST Robotics Club, Chief Engineer (season 2021)
  - FIRST Robotics Club, Alumni Mentor (season 2022-2024)

EXPERIENCE

Shenzhen Loop Area Institute (SLAI) — *Shenzhen, China*

Research Assistant — *Jun 2026 - Present*

- Research on robotics and embodied intelligence.

PROJECT EXPERIENCE

LEGO Assembly Understanding — *Supervised by Prof. Ziqi Wang*

Ongoing Research Project, jointly run by HKUST and SLAI — *Mar 2026 - Present*

- Building a pipeline that automatically generates step-by-step assembly trees for LEGO models: parsing visual assembly manuals into step graphs, then learning a repeat-aware perception policy over a LoRA-adapted DINOv2 backbone to decompose a target model into per-step part subsets, combined with search-based planning.

ELEC5660 Introduction to Aerial Robotics — *Supervised by Prof. Shaojie Shen*

Course Project — *Feb 2026 - May 2026*

- Developed aerial robot algorithms from scratch, including visual-inertial odometry, Extended Kalman Filter (EKF) state estimation, and autonomous path planning within a simulation environment, enabling robust UAV localization, mapping, and navigation.

Quadruped Robot Control — *Supervised by Prof. Maurice Pagnucco and Prof. Yang Song*

Research Practicum — *Feb 2025 - Aug 2025*

- Built and programmed a Unitree GO2 quadruped robot to perform complex tasks, integrating depth camera for accurate object localization (localization error < 10cm) and LiDAR for SLAM.

Underwater Localization System — *Supervised by Prof. Huan Yin and Prof. Fumin Zhang*

Final Year Project — *Jun 2025 - May 2026*

- Developed a Multi-Camera Vision-Based Localization System for Tracking Multiple Autonomous Underwater Vehicles (AUVs). Reduced localization error in complex underwater lighting scenarios, improving positioning accuracy to within 10cm in a $1m \times 1m$ range.

HONORS

2022-2026	**Dean's List,** 5 times	*HKUST*
2023-2026	**Continuing Undergraduate Students Scholarship,** 10,000 HKD per year	*HKUST*
2023	**Bronze Medal,** The Innovation Competition: Our Livable City, We Engineer	*HKIE*
2022	**Entry Scholarship,** 70,000 HKD	*HKUST*

SKILLS

- **C/C++**: Eigen, PCL, STL, STM32, ESP32
- **Python**: PyTorch, OpenCV, Numpy
- **ROS/ROS2**: Navigation2, Gazebo, Slam Toolbox, TF
- **Robotics**: SLAM, VIO, EKF, ICP, Path Planning, Control
- **Deep Learning**: YOLO, ViT, RNN, GAN, VAE, Diffusion, LLM
- **Machine Learning**: GLM, SVM, PCA, XGBoost, HMM
- **Hardware**: Circuit Design, Soldering, CAD, 3D Printing

LANGUAGES

English • Fluent (IELTS 7.5)
Mandarin • Mother tongue